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COMMENTARY

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Improving estimation of flight altitude in wildlife telemetry studies

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Abstract

1. Altitude measurements from wildlife tracking devices, combined with elevation data, are commonly used to estimate the flight altitude of volant animals. However, these data often include measurement error. Understanding this error may improve estimation of flight altitude and benefit applied ecology.
2. There are a number of different approaches that have been used to address this measurement error. These include filtering based on GPS data, filtering based on behaviour of the study species, and use of state-space models to correct measurement error. The effectiveness of these approaches is highly variable.
3. Recent studies have based inference of flight altitude on misunderstandings about avian natural history and technical or analytical tools. In this Commentary, we discuss these misunderstandings and suggest alternative strategies both to resolve some of these issues and to improve estimation of flight altitude. These strategies also can be applied to other measures derived from telemetry data.
4. *Synthesis and applications*. Our Commentary is intended to clarify and improve upon some of the assumptions made when estimating flight altitude and, more broadly, when using GPS telemetry data. We also suggest best practices for identifying flight behaviour, addressing GPS error, and using flight altitudes to estimate collision risk with anthropogenic structures. Addressing the issues we describe would help improve estimates of flight altitude and advance understanding of the treatment of error in wildlife telemetry studies.

KEYWORDS

flight altitude, flight behaviour, GPS, measurement error, model assumptions, state-space models, telemetry studies, wildlife tracking

1 | INTRODUCTION

Telemetry data collected from GPS devices used to track animals are now commonly incorporated into wildlife studies. In addition to latitude and longitude, in the case of volant species, these data usually include information on flight speed, direction, and the altitude above some reference coordinate system. Altitude data can then be converted into an estimate of flight altitude above ground level. Understanding flight altitude has important applied ecological relevance, especially when predicting collision risk with wind turbines, buildings, aircraft, or power lines.

Calculating flight altitude from wildlife telemetry data generally involves subtracting the ground elevation in a digital elevation model (DEM) from the GPS-measured altitude above the same reference surface used to create the DEM (usually either a geoid or an ellipsoid; Miller et al., 2014; Poessel, Brandt, Miller, & Katzner, 2018). This estimate of

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flight altitude usually includes measurement error (sometimes called "observation error"; Hooten, Johnson, McClintock, & Morales, 2017) which accumulates from compounding of errors (1) by the GPS in calculating altitude above the reference surface, (2) by the GPS in calculating latitude and longitude, which can result in inaccurate association of the GPS location with a ground elevation value, and (3) in the ground elevation data used in the DEM, which is caused by inaccurate elevation estimation or by averaging a range of elevations within a single pixel of the DEM.

Accounting for this measurement error is important to improve understanding of flight altitude of birds. This has been done in a number of different ways in the past. Recently, state-space models have been fitted to flight altitude[1] data collected from GPS telemetry units placed on birds (Péron et al., 2017; Ross-Smith et al., 2016). These models separate the variance in flight altitude data into two components described as process (actual vertical movements) error and sampling (measurement or observation) error, and then adjust the flight altitude values by correcting for the measurement error. Adjusted flight altitudes are then used to inform risk to these species from collision with wind turbines, which is an important cause of fatality for many volant species (Loss, Will, & Marra, 2013). Using statistical tools to separate measurement and process error is potentially an improvement over previous work in this field.

Despite potential improvements, however, some studies using telemetry data to estimate flight altitude incorporate erroneous assumptions about avian natural history and technical applications of data, both of which have the potential to weaken inference. Here, we discuss conceptual and technical issues in these studies that, if addressed, would improve estimation of flight altitude from telemetry data. In particular, we (1) identify assumptions about avian natural history sometimes used to filter flight and non-flight behaviour, and we suggest best practices to avoid pitfalls associated with this method, (2) identify technical aspects of collection and processing of GPS data that are essential to understand if one is to correctly interpret those data prior to statistical modelling, and (3) note problems that can occur when state-space models are used to interpret risk. Our goal in highlighting these issues is to advance understanding of the treatment of error and improve estimates of flight altitude in studies that use wildlife telemetry data.

2 | CHARACTERISING FLIGHT VS. NON-FLIGHT BEHAVIOUR

One of the more challenging aspects of the estimation of flight altitude is the first step in processing telemetry data—determining whether a GPS location is indicative of the animal being "in flight". In the study by Péron et al. (2017), Andean condors *Vultur gryphus* and griffon vultures *Gyps fulvus* were assumed to always be in flight between mid-late morning and mid-late afternoon. For a third species, golden eagles *Aquila chrysaetos*, locations were characterised as "in flight" based on inference about distance between two GPS points (i.e. horizontal displacement of >15 m). In contrast, Ross-Smith et al. (2016) collected telemetry data from lesser black-backed gulls *Larus fuscus* and great skuas *Stercorarius skua* and assigned "states" to each GPS location based on the speed of that location. In this last study, locations with a GPS-recorded speed >4 km/hr were classified as being "in flight" and all other locations were assigned to one of three non-flight states.

In all three of these approaches, characterisation of "flight" vs. "non-flight" was based on interpreting a single, potentially flawed, data type. Furthermore, the approaches used by Péron et al. (2017) were based on natural history assumptions that, although they seem reasonable at face value, may be inappropriate. These measurement errors and incorrect natural history assumptions can lead to inaccurate characterisation of flight behaviour.

2.1 | Estimating flight behaviour by time of day

Using time of day to determine flight behaviour assumes strong relationships between probability of being in flight and time of day. It is intuitively correct to assume that large soaring raptors are likely to be in flight at mid-day. However, and perhaps counterintuitively, these same species also are relatively more likely to be feeding on the ground in the middle of the day than they are during any other time. For a large soaring raptor, the decision to land is an important one because, once on flat ground, if updraft is insufficient or wind conditions unsuitable, the bird may not be able to take flight again. For a condor or vulture, being grounded is inherently risky and something to avoid (Horvitz et al., 2014). Thus, a large, soaring raptor is most likely to land to feed when conditions also are most suitable for take-off and flight. Those conditions occur most frequently in the middle of the day (Yates, 1953). Consequently, understanding natural history of large soaring birds reveals that it is not appropriate to assume that they are always flying at mid-day.

To illustrate this point, we evaluated publicly available telemetry data collected from California condors *Gymnogyps californianus*. California condors are closely related to Andean condors, are convergent with griffon vultures, and have substantial behavioural similarities to both species. We filtered these data to remove GPS locations before the birds left their night-time roosts in the morning, perches within 1 km of their night-time roosts, and after they returned to those roosts in the evening. Thus, in this initial round of analyses, we only focused on data collected when birds were >1 km away from their night-time roosts. During this period, estimates of flight altitude (the GPS-recorded altitude minus the DEM value at the GPS-recorded latitude and longitude) suggested that visits to the ground were most likely between 11.00 and 14.00 hr and were less frequent in the early morning and late evening (Figure 1). In a separate analysis, we considered one full year of condor telemetry data that we did not filter by time of day. Here, we found that between 11.00 and 15.00 hr, condors spent, on average, 47% of each hour on the ground (Figure 2; in both cases, the approach we used to characterise GPS data is described in the figure captions). Although we did

[1]The term "flight height" is often used to describe altitude above ground level. The words "altitude" and "height" are, in many regards, synonyms. However, "altitude" is the conventional term used in aeronautics and avian ecology (e.g. Shannon, Young, Yates, Fuller, & Seegar, 2002) to describe a flying object. In general, "altitude" refers to the difference between a point and a fixed reference coordinate system, whereas "height" refers to the distance between two points. For these reasons, we recommend, and use here, the term "altitude" instead of "height" to describe the distance between a flying object and a fixed coordinate system.

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not use state-space models to derive estimates of flight behaviour, the overall pattern these data illustrate is strong, consistent across individuals, and unlikely to change if we used a different method to characterise flight behaviour.

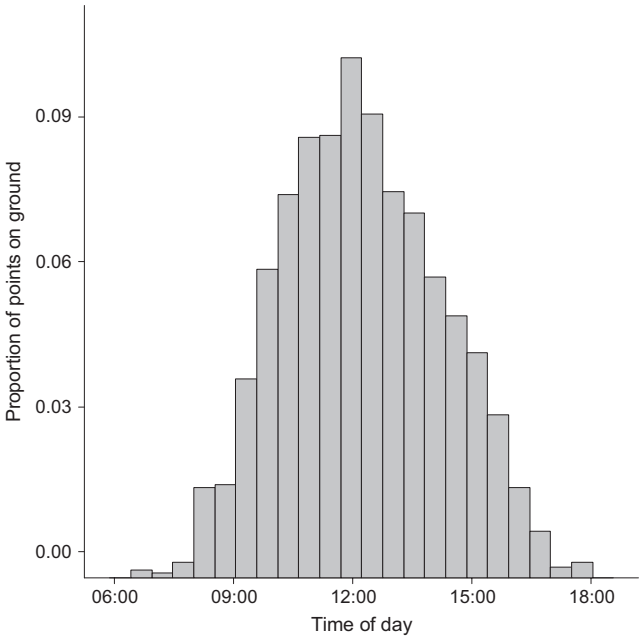


FIGURE 1 Proportion of visits to the ground (n = 1,862) by each half-hour segment of the day for 28 California condors in southern California, 2013–2015. Data were considered only if they were after the condor's first sustained flight of the day, before they roosted at the end of the day, and >1 km from those roost locations and from known nesting sites. Thus, a large majority of the data points in this figure are likely feeding events. Birds were identified as "on the ground" when both their flight speed was ≤3 knots and altitude above ground level was <50 m. Habitat used by California condors ranges from rugged cliffs and mountains to valleys and deserts

These two datasets demonstrate a potentially counterintuitive natural history characteristic of large soaring birds. That is, the more likely scavenging birds are to take advantage of optimal flying conditions and fly (Figure 2), the more likely they are to search for, find, and utilise food resources on the ground (Figure 1). Thus, it is likely incorrect to assume that large soaring birds are always "in flight" based on time of day.

Furthermore, truncating these data to only part of the daylight hours (as Péron et al. 2017 did) has important consequences for understanding collision risk; a model that considers a part of the day can only evaluate risk during that period. As our data illustrate, in the *c.* 50% of daylight hours before 11.00 hr and after 15.00 hr, condors still spent a great deal of their time flying. Ignoring these data can cause incorrect inference, especially when flight behaviour, and thus flight altitude and collision risk, changes with time of day. Specific to this situation, flight altitudes of soaring birds generally are lower in morning and evening (Katzner et al., 2015; Poessel et al., in press), and thus a model of low-altitude collision risk without those times of the day will produce a misunderstanding of the actual threats that birds face. Thus, using time of day to infer condor or vulture flight behaviour is an oversimplification that not only affects classification of that flight behaviour, but also influences our understanding of altitude-dependent collision risk.

2.2 | Estimating flight behaviour by distance moved

When using distance moved to determine flight behaviour, inference also is weak. Horizontal displacement over a long distance only indicates that a bird moved between two points. It reveals nothing about the bird's behaviour at any instant between and including the times when GPS positions are calculated. Thus, a bird may fly far in the period between two consecutive locations but may still be perched when GPS data are collected at either end of that period.

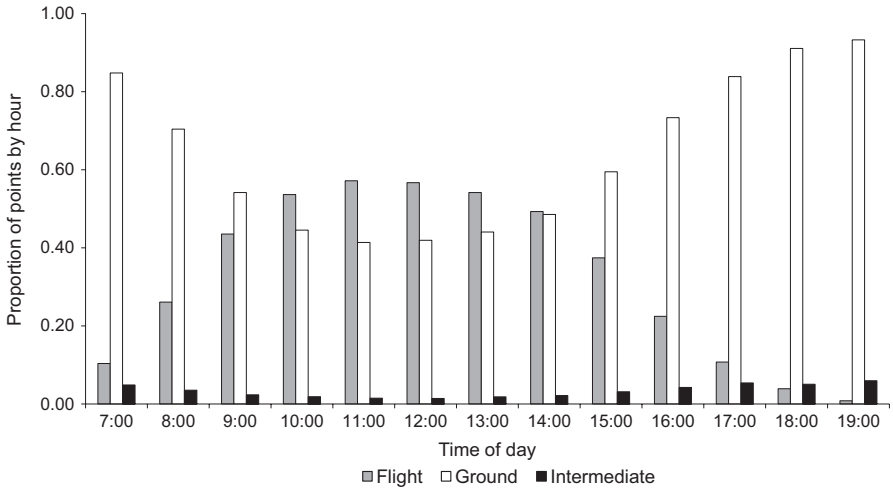


FIGURE 2 Proportion of GPS locations classified as being in flight, on the ground, or intermediate by hour of day between 07.00 and 19.00 hr for 26 California condors in southern California, 2015. This figure shows the hourly probability of condors being in each of the three states, and the pattern appears opposite from that in Figure 1, which shows temporal patterns in ground points only. Condors were characterised as "in flight" when speed was ≥4 knots, "on the ground" when both speed was ≤3 knots and flight altitude was <50 m, and "intermediate" when speed was >3 knots and <4 knots and also when both speed was ≤3 knots and flight altitude was ≥50 m

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Alternatively, a bird soaring in a thermal may travel many hundreds of metres vertically between two consecutive GPS locations, yet its horizontal displacement could be <15 m. Finally, some birds, such as golden eagles and vultures, tend to walk long distances, and they can easily walk >15 m in a very short time interval. These realities apply to behavioural classifications regardless of the time interval between GPS fixes.

To illustrate the problem of using horizontal displacement to define flight, we evaluated GPS telemetry data collected at 15-min intervals from five resident golden eagles tracked in the Mojave Desert of California (described in Braham et al., 2015). For those data, estimates of flight altitude suggested that 59% of GPS telemetry locations that were ≥15 m from the previous location were measurements of perched birds (Figure 3; the approach we used to characterise GPS data is described in the figure caption). Thus, using horizontal displacement of GPS locations to infer flight behaviour also is another oversimplification that can lead to incorrect classification of that flight behaviour.

2.3 | Best practices for identifying behaviour as "in flight"

There is no perfect way to characterise a GPS data point as indicative of any behavioural state. That said, our review of prior work suggests a few best practices that could be applied to flight altitude studies. First, any characterisation of behaviour should be based on a firm understanding of the natural history of the animal in question. Plots of uncorrected raw data may help determine which assumptions are and are not plausible. In the previously noted studies, such plots would have indicated a large number of data points on the ground at mid-day. Second, it is often helpful to use two different GPS measurements, rather than just one, to estimate flight behaviour. This approach removes circularity inherent in modelling a characteristic of a dataset that has been parsed by that characteristic (although this is done regularly, such as when modelling speed of a moving animal and ignoring times when the animal is not moving). For example, in our work here and in papers we and others have published (Katzner et al., 2012; Miller et al., 2014; Poessel, Brandt, et al., 2018; Shamoun-Baranes, Bouten, Camphuysen, & Baaij, 2011), flight behaviour was interpreted based on GPS data describing both instantaneous speed and altitude above ground level. In theory, information on time of day and horizontal displacement could be incorporated in a probabilistic manner in conjunction with measurements of speed and altitude (although with the caveats of our discussion above). Finally, characterisation of behaviour may be improved by state-space or other models that consider an entire dataset, not a temporally truncated one. Although such data are likely more difficult to model, considering the entire dataset would be an improvement on any of the truncation approaches described above.

3 | TECHNICAL ASSUMPTIONS

In the era of "big data," machines do the majority of the data collection once performed by people. Although this is undoubtedly a good thing, the trend also creates additional responsibility for researchers to understand how their data are collected and the assumptions that underlie these data. Misunderstanding these technical processes has weakened some prior studies, but understanding them will improve the estimation of flight altitude and other biology based on GPS data. We discuss these issues, including (1) manufacturer-specific and point-specific variation in measured GPS error, (2) negative flight altitude values, and (3) differences between the reference surfaces used by the GPS and those used in estimation of the DEM, and the consequences of those differences in estimating flight altitude.

3.1 | GPS error

Not all telemetry units are equal. Variation in GPS fix quality (precision and accuracy) is due to variation in commercially available GPS modules (the components inside the tracking device), to the size and quality of the GPS antenna, and to how these "off-the-shelf" components are integrated by a manufacturer into a telemetry device. Furthermore, some telemetry device manufacturers upgrade hardware and software as improvements become available, and thus GPS

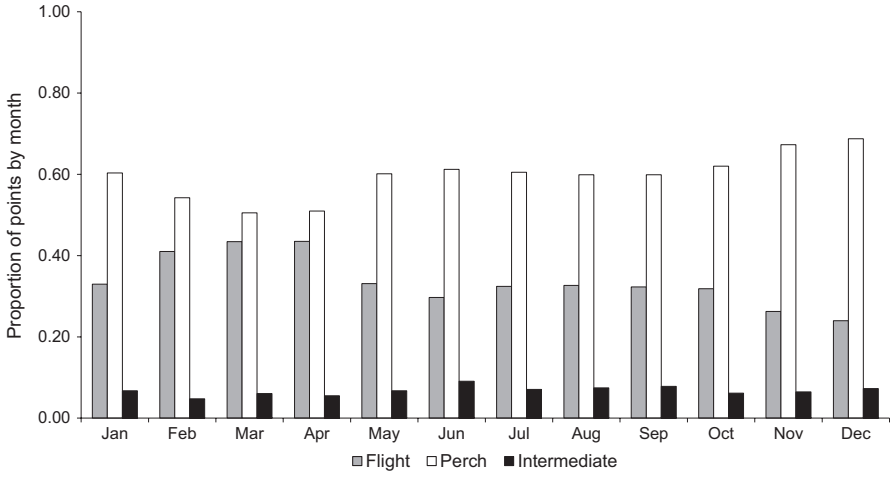


FIGURE 3 Proportion of GPS locations by month that were ≥15 m from the previous location and that were classified as being in flight, perched, or intermediate between 08.00 and 17.00 hr for five golden eagles in the Mojave Desert of California, 2012–2013. Eagles were characterised as "in flight" when speed was >4 knots, "perched" when both speed was ≤4 knots and flight altitude was ≤50 m, and "intermediate" when both speed was ≤4 knots and flight altitude was >50 m

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fix quality changes with time and among manufacturers (Blackie, 2010). Finally, GPS telemetry devices produced by some manufacturers can have an error that irregularly truncates altitude data (e.g. altitude is reported as 100 m when it is actually 1,000 m; authors' unpublished observations). Although none of the studies mentioned previously use GPS devices produced by different manufacturers in a single statistical model, many studies incorporate tracking data from different makes or models of telemetry devices, and in such cases, inter- and intra-manufacturer variations are important to consider.

Understanding variation among devices is important when interpreting variation in fix quality by individual telemetry units. Errors in fix quality have been addressed by filtering GPS telemetry locations by the number of satellites used to attain the fix (e.g. Péron et al., 2017 considered only GPS data for eight vultures obtained when ≥6 satellites were available). However, number of satellites may not always be an adequate measure of fix quality. This is because satellite position also influences measurement accuracy. A better metric to use with GPS data is dilution of precision (DOP; generally in either horizontal [HDOP] or vertical [VDOP] planes). Measures of DOP are useful in estimating precision because they incorporate in their calculations not only the number of satellites, but also the position of those satellites relative to the GPS device (their geometry; Theobald, 2007). Thus, several satellites close together will provide poor locational precision, resulting in higher DOP values. In contrast, when satellites are farther apart, their geometry allows improved triangulation and fix quality, resulting in lower DOP values. Furthermore, the distribution of errors of GPS fixes is dependent on fix quality (i.e. the magnitude and distribution of error of points with HDOP <4 is almost certain to be very different from the magnitude and distribution of error of points with HDOP >10). As a consequence, although DOP is rarely used in research, it is provided by many modern telemetry devices and it is an essential parameter to incorporate when filtering data.

Ignoring DOP values can result in substantial error both in the assumptions made about animal behaviours used to filter data and in the statistical models used to analyse those data. Ross-Smith et al. (2016) incorporated DOP as a linear effect into state-space models showing that variation in DOP was strongly related to variation in flight altitude error. That error is also relevant when considering Péron et al.'s (2017) 15-m movement threshold for golden eagles. This is because GPS error on points with HDOP >5 is usually, by definition, >15 m (this is "by definition" because many modern GPS units have a measurement standard deviation [*SD*] of approximately ±3 m, and position *SD* is approximated by multiplying DOP by the measurement *SD*). Thus, without accounting for the error in quality of individual fixes, two points may appear >15 m apart either because an animal moved or simply because of measurement error.

3.2 | Negative flight altitude values

Even when DOP is taken into account, once an animal is determined to be in flight, some records will suggest that a flying animal is below the ground surface. The causes of these incorrect flight altitudes are from the compounding of errors in the elevation data (the DEM) with GPS errors that place the bird at the incorrect location in the horizontal and vertical planes. The simplest approach to address these negative flight altitude records is to remove locations below some threshold. For example, Katzner et al. (2012) removed altitudes <−50 m above ground level, because this is approximately the sum of the GPS error (*c.* 15 m), error in the DEM (*c.* 3 m), and error in interpolation (*c.* 30 m). In contrast, Péron et al. (2017) fitted a continuous-time stochastic movement model within a state-space framework to "filter out" the measurement error from flight altitude data. Finally, Ross-Smith et al. (2016) developed similar models that incorporated negative observations, but that ensured that all estimated true altitude values were above sea level (ASL; positive altitudes). Although the incorporation of flight altitude values into state-space models is more complex than the use of a threshold, such models have their own weaknesses that may limit their utility and may affect ecological inference (Auger-Méthé et al., 2016). As such, improving the analytical tools to address negative altitude values is an important step in improving understanding of flight behaviour.

3.3 | Altitude data measured above the ellipsoid

Altitude and ground elevation are usually estimated in reference to one of two approximations of the earth's surface. The ellipsoid is a mathematical model that approximates the three-dimensional shape of the earth. In contrast, mean sea level is the zero elevation for a local area and is a response to the earth's gravitational field (Fraczek, 2003).

The difference between altitude above the ellipsoid and altitude ASL can be as great as 105 m (e.g. near India; Fraczek, 2003). Since nearly all modern horizontal-axis wind turbines are <150 m in height, such a difference is obviously relevant when evaluating risk to flying birds from wind turbines. Thus, if a particular study uses data from two different sources (e.g. a DEM that measures ground elevation ASL and telemetry units that collect altitude data as metres above the ellipsoid), it is critical to integrate data appropriately. To calculate accurate flight altitudes above ground, altitude measurements collected in reference to the ellipsoid must be converted to altitude ASL. Only then can ground elevation ASL (the approximation generally used in a DEM) be subtracted from altitude ASL to estimate flight altitude above ground level. Some telemetry manufacturers perform this conversion for the user. In other cases, the user must perform this conversion on their own. For interested users, we outline a series of steps to perform this operation (Appendix S1).

Péron et al. (2017) state that the high rate of error that they found (i.e. 36% of their flight altitude records were classified as underground) is typical when flight altitude is calculated and that their state-space model is an attempt to correct for this error. We do not see such large errors as typical, and using our filtering approach based on two simultaneous GPS measurements, our research team has not reported errors of this magnitude. Such a large error rate could occur if two sets of measurements (GPS-derived altitude and ground elevation) were collected against two different frames of geodetic reference. Best practices for accurate estimation of flight altitude require comparison against a single reference surface (i.e.

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altitude ASL vs. ground elevation ASL or, alternatively, altitude above the ellipsoid vs. ground elevation above the ellipsoid). Comparing data collected against two different reference surfaces (e.g. altitude above the ellipsoid vs. ground elevation ASL) will automatically induce measurement error and undermine inference.

4 | FROM FLIGHT ALTITUDE TO RISK MODELLING

Finally, the methodology used to adjust flight altitude values and subsequently estimate collision risk must be applied with caution. For example, the methodology used by Péron et al. (2017) appears to be problematic because of circularity. The authors' two-step process includes first adjusting flight altitude by using a state-space model to separate process error from measurement error. Process error in flight altitude was modelled as a log-linear relationship with thermal updraft potential, such that higher values of flight altitude were associated with higher values of thermal updraft potential and higher levels of process error. Inherently, lower values of flight altitude were then related to lower values of thermal updraft potential. The second step of their process was to use the adjusted flight altitude as a response variable to determine which of several factors influenced risk of birds colliding with wind turbines. A key variable that they found explained risk was thermal updraft potential. This is not at all surprising because that same variable was used to adjust flight altitude (the variable used to calculate risk). This is circular logic because, in the second step of their analyses, both the predictor and response variables were a function of thermal updraft potential.

In a more general context, when collision risk is estimated by modelling flight altitudes, any given variable (e.g. thermal updraft potential) should not be used to both adjust flight altitude (a measure of risk) and determine factors that influence risk. For example, adjusted flight altitude is not only correlated directly with thermal updraft potential, but also with surface sensible heat flux and temperature, which are used to calculate thermal strength (Bohrer et al., 2012). Thus, if environmental variables must be used to both correct the response variable and act as predictor variables, those used as predictors should be different from those used to affect responses. In fact, there are a suite of variables that correlate with flight altitude of soaring birds (e.g. see Poessel et al., in press) but are uncorrelated with each other. Use of these variables would be logically consistent for statistical modelling.

5 | FINAL THOUGHTS

Improving treatment of error in flight altitude data in telemetry studies of birds is an important goal that has the potential to improve understanding of animal flight and collision risk. Although not all of the issues we describe are equally consequential, each alone is significant enough to change interpretation of risk and behaviour. Thus, effectively interpreting telemetry data requires not only the scientific and technical skills to analyse these data, but also a well-grounded understanding both of the natural history of the animals being studied and of modern telemetry devices and mapping datasets. Simultaneously grasping all these diverse components is not trivial for any ecologist. We hope that raising the issues we present here will advance understanding of the treatment of error in studies that use wildlife telemetry data, to the ultimate benefit of science and conservation.

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AUTHORS' CONTRIBUTIONS

S.A.P. and T.E.K. conceived of the rationale for this study and led the writing. Each author wrote a section of the manuscript, and S.A.P. and T.E.K. led integration of these sections. J.C.H. prepared the California condor information and data in Figure 1, and M.A.B. prepared Figure 2 and the golden eagle data in Figure 3. All authors contributed to revising and improving the manuscript and gave final approval for publication.

DATA ACCESSIBILITY

Telemetry data for California condors are publicly available from ScienceBase Catalog, item 53276178e4b00296c00bbed5 (Poessel, Duerr, Hall, Braham, & Katzner, 2018). Specific "Active data" file names: "gsm-sCal-2014-10-21", "gsm-sCal-2014-12-29", "gsm-sCal-2015-03-31", "gsm-sCal-2015-06-30", "gsm-sCal-2015-09-30", and "gsm-sCal-2015-12-31".

Telemetry data for golden eagles are sensitive and have not been archived.

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